

Hongjie Fang

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Education

Shanghai Jiao Tong University

Ph.D. Student, Wu Wenjun Honorable Class, Computer Science and Engineering

2022/09 – present

Shanghai, China

Shanghai Jiao Tong University

Bachelor of Engineering, major in Computer Science and Engineering,

2018/09 – 2022/06

Bachelor of Economics, minor in Finance

Shanghai, China

- GPA 4.03 / 4.3, Ranking: 2 / 149.

Awards

[Best Paper Award](#), ICRA 2024

2024

[Best Paper Award](#), IROS 2023 Manipulation and Grasping Workshop

2023

Shanghai Outstanding Graduates

2022

Shanghai Scholarship

2020

National Olympics in Informatics (NOI) Bronze Medal

2017

National Olympics in Informatics in Provinces (NOIP) First Prize

2015 & 2016

Publications

1. AirExo-2: Scaling up Generalizable Robotic Imitation Learning with Low-Cost Exoskeletons
Hongjie Fang*, Chenxi Wang*, Yiming Wang*, Jingjing Chen*, Shangning Xia, Jun Lv, Zihao He, Xiyan Yi, Yunhan Guo, Xinyu Zhan, Lixin Yang, Weiming Wang, Cewu Lu, Hao-Shu Fang
Conference on Robot Learning (CoRL), 2025, [\(Oral Presentation\)](#).
2. AirExo: Low-Cost Exoskeletons for Learning Whole-Arm Manipulation in the Wild
Hongjie Fang*, Hao-Shu Fang*, Yiming Wang*, Jieji Ren, Jingjing Chen, Ruo Zhang, Weiming Wang, Cewu Lu
IEEE International Conference on Robotics and Automation (ICRA), 2024.
3. TransCG: A Large-Scale Real-World Dataset for Transparent Object Depth Completion and A Grasping Baseline
Hongjie Fang, Hao-Shu Fang, Sheng Xu, Cewu Lu
IEEE Robotics and Automation Letters (RA-L), 2022; Presented at **ICRA** 2023.
4. History-Aware Visuomotor Policy Learning via Point Tracking
Jingjing Chen*, **Hongjie Fang***, Chenxi Wang, Shiquan Wang, Cewu Lu
IEEE International Conference on Robotics and Automation (ICRA), 2026.
5. FoAR: Force-Aware Reactive Policy for Contact-Rich Robotic Manipulation
Zihao He*, **Hongjie Fang***, Jingjing Chen, Hao-Shu Fang, Cewu Lu
IEEE Robotics and Automation Letters (RA-L), 2025; Presented at **IROS** 2025.
6. Learning Dexterous Manipulation with Quantized Hand State
Ying Feng*, **Hongjie Fang***, Yinong He*, Jingjing Chen, Chenxi Wang, Zihao He, Ruonan Liu, Cewu Lu
IEEE International Conference on Robotics and Automation (ICRA), 2026.
7. RH20T: A Comprehensive Robotic Dataset for Learning Diverse Skills in One-Shot
Hao-Shu Fang, **Hongjie Fang**, Zhenyu Tang, Jirong Liu, Chenxi Wang, Junbo Wang, Haoyi Zhu, Cewu Lu
IEEE International Conference on Robotics and Automation (ICRA), 2024
8. RISE: 3D Perception Makes Real-World Robot Imitation Simple and Effective
Chenxi Wang, **Hongjie Fang**, Hao-Shu Fang, Cewu Lu
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2024.
9. CAGE: Causal Attention Enables Data-Efficient Generalizable Robotic Manipulation
Shangning Xia, **Hongjie Fang**, Hao-Shu Fang, Cewu Lu
IEEE International Conference on Robotics and Automation (ICRA), 2025.

10. Towards Effective Utilization of Mixed-Quality Demonstrations in Robotic Manipulation via Segment-Level Selection and Optimization
Jingjing Chen, **Hongjie Fang**, Hao-Shu Fang, Cewu Lu
IEEE International Conference on Robotics and Automation (ICRA), 2025.
11. AnyGrasp: Robust and Efficient Grasp Perception in Spatial and Temporal Domains
Hao-Shu Fang, Chenxi Wang, **Hongjie Fang**, Minghao Gou, Jirong Liu, Hengxu Yan, Wenhai Liu, Yichen Xie, Cewu Lu
IEEE Transaction on Robotics (T-RO), 2023; Presented at ICRA 2024, (**Best Paper Award on IROS 2023 Workshop**).
12. Motion Before Action: Diffusing Object Motion as Manipulation Condition
Yue Su*, Xinyu Zhan*, **Hongjie Fang**, Yong-Lu Li, Cewu Lu, Lixin Yang
IEEE Robotics and Automation Letters (RA-L), 2025; Presented at ICRA 2026.
13. Dense Policy: Bidirectional Autoregressive Learning of Actions
Yue Su*, Xinyu Zhan*, **Hongjie Fang**, Han Xue, Hao-Shu Fang, Yong-Lu Li, Cewu Lu, Lixin Yang
IEEE/CVF International Conference on Computer Vision (ICCV), 2025.
14. Knowledge-Driven Imitation Learning: Enabling Generalization Across Diverse Conditions
Zhuochen Miao*, Jun Lv*, **Hongjie Fang**, Yang Jin, Cewu Lu
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2025.
15. Open X-Embodiment: Robotic Learning Datasets and RT-X Models
Open X-Embodiment Collaboration, 147 authors
IEEE International Conference on Robotics and Automation (ICRA), 2024, (**Best Paper Award**).
16. Flexible Handover with Real-Time Robust Dynamic Grasp Trajectory Generation
Gu Zhang, Hao-Shu Fang, **Hongjie Fang**, Cewu Lu
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2023.
17. SIME: Enhancing Policy Self-Improvement with Modal-Level Exploration
Yang Jin*, Jun Lv*, Wenye Yu, **Hongjie Fang**, Yong-Lu Li, Cewu Lu
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2025.
18. Graspness Discovery in Clutters for Fast and Accurate Grasp Detection
Chenxi Wang*, Hao-Shu Fang*, Minghao Gou, **Hongjie Fang**, Jin Gao, Cewu Lu
IEEE/CVF International Conference on Computer Vision (ICCV), 2021.
19. Target-Referenced Reactive Grasping for Dynamic Objects
Jirong Liu, Ruo Zhang, Hao-Shu Fang, Minghao Gou, **Hongjie Fang**, Chenxi Wang, Sheng Xu, Hengxu Yan, Cewu Lu
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023.

Preprints

1. AnyDexGrasp: Learning General Dexterous Grasping for Any Hands with Human-level Learning Efficiency
Hao-Shu Fang, Hengxu Yan, Zhenyu Tang, **Hongjie Fang**, Chenxi Wang, Cewu Lu
arXiv, 2025.

Academic Services

Reviewer for journal *RA-L*, *T-CYB*, *T-MECH*.

Reviewer for conferences including *ICRA* (2023 - 2026), *IROS* (2023 - 2025), *CoRL* (2025), *NeurIPS* (2025), *ICLR* (2025), *etc.*

Invited Talks

- 03/2024, **Echo AI @ USyd**, Towards Efficient Robot Imitation Learning from Human Demonstrations.
- 11/2024, **Zhixingxing**, Towards Efficient Robot Imitation Learning from Human Demonstrations.
- 02/2025, **Yang Gao Group @ THU**, Towards Generalizable Imitation Learning from Human Demonstrations.
- 08/2025, **3D CVer**, Towards Generalizable Imitation Learning from Human Demonstrations.
- 08/2025, **Sharpa**, Towards Generalizable Imitation via Scalable Data and Robust Policy.
- 09/2025, **Galbot**, Building Capable and Generalizable Policies for Robotic Manipulation.
- 09/2025, **Shenlanxueyuan**, Towards Generalizable Imitation via Scalable Data and Robust Policy.

- 10/2025, **The Robot Dexterity Lab @ Duke**, *AirExo-2: Scaling up Generalizable Robotic Imitation Learning with Low-Cost Exoskeletons*.

Teaching

Teaching Assistant, <i>Algorithm and Complexity</i>	<i>Spring, 2022</i>
Teaching Assistant, <i>C++ Programming Language (Honor)</i>	<i>Fall, 2020</i>
Teaching Assistant, <i>Data Structure (Honor)</i>	<i>Spring, 2019</i>
Teaching Assistant, <i>Linear Algebra (Honor)</i>	<i>Fall 2021 & Fall, 2022</i>
Teaching Assistant, <i>Mathematical Analysis (Honor)</i>	<i>Fall, 2020 & Fall, 2021</i>